Journal of Chromatography A, 1609 (2020) 460440

Contents lists available at ScienceDirect

Journal of Chromatography A

journal homepage: www.elsevier.com/locate/chroma

Critical parameters of liquid chromatography at critical conditions in context of poloxamers: Pore diameter, mobile phase composition, temperature and gradients

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ARTICLE INFO

Article history:
Received 12 June 2019
Received in revised form 6 August 2019
Accepted 7 August 2019
Available online 8 August 2019

Keywords:
Block copolymers
Liquid chromatography at critical conditions (LCCC)
Oligomers
Chromatographic critical Point (CCP)
Interaction parameter

ABSTRACT

At the borderline between size exclusion chromatography (SEC) and interaction chromatography (IC) there is a special mobile phase composition and temperature at which polymer chains become "chromatographically invisible". This point is termed as "chromatographic critical point" and chromatographic separations performed using these conditions are called "liquid chromatography at critical conditions" (LCCC). LCCC is a powerful technique in the analysis of functional polymers and block copolymers. At these so-called critical conditions molar mass discrimination of any specific homopolymer is suppressed rendering elution of whole range of molar mass at same elution volume. These conditions allow enhanced separation with regard to non-critical segment either in exclusion or interaction regime of the polymer chromatography. This article is intended to critically discuss different parameters that can be maneuvered to improve separation and in turn characterization of non-critical segment of block copolymers at LCCC. Different experimental parameters evaluated in this study include pore size of the column, mobile phase composition, temperature and gradients. These parameters can be adeptly adjusted to improve separation of non-critical segment while keeping the other segment close to critical conditions. Current study demonstrates that pore diameter and mobile phase are the only practical variable that can be used for improvement of characterization of non-critical block in the block copolymer while non-critical block is in exclusion regime. On the other hand, pore diameter of the column, temperature, solvent composition and gradients are important parameters that can be skillfully tuned for improvement of separation of non-critical block while non-critical block elutes in interaction regime. The above-mentioned variations are evaluated for di-block as well as tri-block copolymers of A-B-A and B-A-B type. Moreover, LCCC-IC is especially important for analysis of poloxamers.

1. Introduction

Liquid chromatography at critical conditions (LCCC) is a powerful technique in polymer analysis [1–5]. LCCC has been used for separating polymer blends and functional homopolymers with respect to number and nature of the end-groups [6–9]. Furthermore, LCCC is the only technique for estimation of individual block lengths of the block copolymers along with information with regard to presence of unwanted homopolymers in the sample [1–3,10,11]. The separation of targeted product is controlled by molar mass of non-critical segment of the complex polymeric system while critical homopolymers elute at the critical point near the void volume of the column. The non-critical block may either be excluded from the stationary phase eluting earlier than the critical homopolymers [9–16] or interact more strongly with the stationary phase eluting later than the critical homopolymers [9,17–24]. The accuracy with regard to estimation of molar mass of non-critical block of block copolymers is sometime questioned [25,26]. Another problem associated with LCCC is peak broadening and limited recovery of high molar mass critical homopolymers [27–33]. Nonetheless, LCCC is the only available method for analysis of individual block lengths of the block copolymers and obtained results represent a fairly good estimation if not exact [11,14,34–36].

Basically, the chromatographic critical conditions in polymer analysis refer to the concept of "chromatographic invisibility" of the critical polymer with regard to whole range of molar mass. The phenomenon renders to elution of whole molar mass range at the same elution volume termed as "chromatographic critical point". The first impression of the idea seems not to be very use-

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https://doi.org/10.1016/j.chroma.2019.460440

ful since chromatography being a separation technique and this approach suppresses molar mass selectivity of the separation system. Nonetheless, the concept is genial that suppresses selectivity of the critical segment while magnifying the selectivity of separation with regard to the non-critical segments. This is actually the only technique that allows fair estimation of individual block length of non-critical block of the block copolymers [11,14,34–36]. The idea was first presented by Belenkii and Gankina [2]. The chromatographic critical conditions for any polymer are generally established on typical HPLC reversed and normal phase columns by varying the mobile phase composition and temperature. If the mobile phase is too strong for the polymers and hinders any interaction of repeat unit with the stationary phase, the polymer molecules get separated in the pore volume of the column. The pore sizes vary in their diameter and length that impede access of polymer molecules in them. The larger molecules have less access to the pores and elute earlier whereas smaller molecules have more access to the pores and elute later. However, whole molar mass range in SEC regime must elute earlier than the void volume of the column. SEC is an entropy driven process. Conversely, when analyte has more compatibility with the stationary phase compared to the mobile phase, it interacts with stationary phase rendering to elution of homopolymers in order of increasing molar mass after the void volume of the column. Similar to LC of small molecules, the interactions between analyte and stationary phase are enthalpy controlled. Hence, the enthalpic and entropic forces are competitive and the dominant force will determine which mode of LC of polymers is operative. At the intersection of these two modes of LC of polymers, there is a point when entropic and enthalpic forces compensate each other resulting in molar mass independent elution of that specific polymer near void volume of the column. This special mode of LC of polymers is termed as liquid chromatography at critical conditions (LCCC) and this point is generally termed as critical adsorption point. However, since chromatographic interactions could be either adsorption or partition based, therefore the term is not real representative of the conditions, hence, a new term chromatographic critical point (CCP) will be used in this manuscript and author recommends the use of term “chromatographic critical point” for this situation, being most representative expression [1–5].

The establishment of chromatographic critical conditions requires a set of narrow molar mass standards of the homopolymer. The homopolymers elute in SEC regime in stronger mobile phase composition. The strength of the mobile phase can be reduced to get homopolymers interact with the stationary phase rendering elution of homopolymers in order of increasing number of repeat after the void volume of the column. In between these two points, there is a mobile phase composition at a given temperature that suppresses the molar mass selectivity of that particular homopolymer. Temperature is often used for fine-tuning of the established near critical conditions [37–42]. An excellent method of estimation of critical mobile phase composition based on gradient elution polymer chromatography has been proposed [43–47]. The authors concluded that the chromatographic critical conditions are generally near to the mobile phase composition at the elution of highest molar mass homopolymer.

Theoretically, the distribution coefficient K in SEC regime of polymer chromatography assumes values between 0-1. Its value is more than 1 in case of interaction chromatography whereas at chromatographic critical point the value of distribution coefficient equals unity. Another term that is also used for describing retention mechanism in chromatography of polymers is interaction parameter, c [48,49]. It is negative in size exclusion, positive in interaction chromatography and has value equal to zero at chromatographic critical conditions [1,2].

Table 1
Possible situations in LCCC of block copolymers.

Column	Mobile phase polarity	Polar segment	Non-polar segment
RP	high	LCCC	IC
RP	low	SEC	LCCC
NP	high	LCCC	SEC
NP	low	IC	LCCC

Analysis of block copolymers by liquid chromatography at critical conditions means that one of the blocks assumes value of interaction parameter equal to zero. However, the other block must assume some positive or negative value that makes the basis of separation. The possibilities of elution of non-critical block with reference to nature of mobile phase and stationary phase are summarized in Table 1. Ideally, the length of the critical block would not influence the elution of the block copolymer and block copolymers would elute as per chromatographic behavior of non-critical block either in exclusion or interaction regime. The chromatographic critical conditions of polar block are possible on RP column in mobile phase of high polarity. Under these conditions, the less polar block would have stronger interaction with the stationary phase that render late elution of the block copolymer after the void volume of the column. On the same column, critical conditions of non-polar segment are possible in mobile phase of rather low polarity rendering exclusion of the polar segments that resulted in earlier elution of block copolymer according to the length of the excluded block. On the other hand, critical conditions of polar block are possible on NP column in mobile phase of high polarity. Under these conditions, non-polar blocks are excluded that allow earlier elution of block copolymer as per length of the non-critical block. On the same lines, critical conditions of non-polar block are possible on NP column in mobile phase of low polarity rendering elution of polar block in interaction regime that dictates the elution of the block copolymer.

Negative value of interaction parameter c of non-critical block results in separation of block copolymer in SEC regime earlier than the elution volume of critical homopolymers with respect to length of the excluded block. This is the most versatile approach of LCCC that has been applied to analysis of a wide range of block copolymers. In this manuscript the exclusion of non-critical segment at critical conditions of other block will be referred as liquid chromatography at critical conditions-size exclusion chromatography (LCCC-SEC). This approach is straight forward in its execution and the only important parameter that needs to be considered is pore diameter of the stationary phase along with critical mobile phase selectivity for the hydrodynamic volume of non-critical block. Smaller pore columns are preferred for low molar mass polymers for better resolution with respect to the molar mass of visible segment compared to high molar mass block copolymers that get better resolved according to the molar mass of the visible segment while using stationary phase having larger pores. A calibration curve constructed using homopolymers of the same chemistry as the non-critical block can then be used for estimation of individual block length of the non-critical block of the block copolymers. Nonetheless, the effect of length of the critical homopolymers while estimating length of non-critical block has to be carefully evaluated. In some cases, the effect is very pronounced leading to over- or under-estimation of the non-critical block length [25,26]. On the other hand, in most of the cases the effect is not much pronounced. After careful evaluation of above factors, a reasonable estimation of non-critical block length is possible by LCCC-SEC [1,2,11,14,34–36].

On the contrary, there are several complications and limitations while applying any LCCC method when the non-critical block or end-group is in interaction regime. The late elution of the block copolymer due to interaction of non-critical block at critical conditions of other blocks will be referred as liquid chromatography at critical conditions-interaction chromatography (LCCC-IC). The applications of LCCC-IC are limited to functional polymers or to block copolymers having shorter non-critical blocks with low positive interaction parameter. The functional homopolymers in this case would elute in order of interaction strength of the end-group with the stationary phase at their critical point independent of molar mass discrimination [6,7]. In case of block copolymer, however, certain tactics can be applied to improve resolution and performance of the technique. It is generally thought that freedom of adjusting the operating conditions in LCCC is very limited. Chromatographic critical conditions generally occur as a sharp point that is very sensitive to changes in mobile phase composition and temperature. Variation of mobile phase composition by less than 1% results in switching of separation mechanism from SEC to IC and vice versa. Moreover, existence of two close critical points for PEO is reported in aqueous methanol and aqueous acetonitrile mobile phases on RP octadecyl stationary phases [17,18]. The elution of different molar masses of PEOs between these two critical mobile phase compositions does not show any significant change and remain fairly close to critical conditions. Hence, the elution of oligomers of non-critical block is not effected to large extent within limits of two critical points.

The value of interaction parameter is also temperature dependent and so does the chromatographic critical point [37–42]. Therefore, temperature can also be used to improve the resolution of block copolymer. The best situation is when interaction parameters of critical and noncritical block move in opposite direction with change in temperature.

In one of our previous studies, we have shown how pore diameter can be used to get resolution of oligomers of EO-PO di- and tri-block copolymers according to noncritical block in LCCC-IC [18]. This is a limited freedom as not many RP columns with different porosity are available. In this study, we demonstrate critical parameters including pore size of the column, temperature, solvent composition and gradients to improve the separation of poloxamers at chromatographic critical conditions by taking examples of different di- and tri-block copolymers and mobile phase systems on octadecyl RP columns of different pore diameters. It is pertinent to mention here that the discussion regarding LCCC-SEC is applicable to all the polymers in general. However, the situation discussed for LCCC-IC is a special situation for couple of reasons. Firstly, it is not common to have combination of critical conditions for one block and interaction parameter of interacting block in the range that allows its oligomer separation. Secondly, the existence of two close critical points is known only for PEO on RP octaedecyl columns in aqueous acetonitrile and aqueous methanol mobile phases. The mobile phase composition between these two critical points does not affect the elution to large extent and PEOs show fairly critical behavior. Furthermore, the interaction parameter of interacting block at CCP for PEO should be in a range that allows oligomer resolution. The above-mentioned conditions are fulfilled by commercially important polymers called poloxamers consisting of EO and PO blocks. Hence, the discussed parameters are applicable to poloxamers or for the low molar mass block copolymers of PEO with a suitable combination of interaction strength of the interacting block. Moreover, gradients of two critical mobile phases are applied to di- as well as tri-block copolymers of A-B-A and B-A-B type. In this manuscript, we demonstrate the selectivity of above-mentioned variables for improvement of separation of non-critical block both in LCCC-SEC and LCCC-IC.

2. Experimental part

2.1. Materials

Homopolymer standards of varying molar masses of PEG, PPG, PMMA were purchased from Aldrich (Germany) and Polymer Standards Service (Mainz, Germany), respectively. The block copolymers used for current study are either synthesized in our labs or purchased from the manufacturer. Commercial Poloxamers (EO-PO-EO) were purchased from BASF. Di- and tri-block copolymers of poly(ethylene oxide) (PEO) with poly(methyl methacrylate) (PMMA) were synthesized by ATRP of methyl methacrylate using PEO and methoxy poly(ethylene oxide) (MeO-PEO) of different molar masses as macro-initiator [13]. Di- and tri-block copolymers of ethylene oxide with propylene oxide were synthesized according to procedure reported in one of our previous article [50]. All the block copolymer samples are synthesized by taking equal mass ratio of macro-initiator to monomer and polymerizing until all the monomer reacted. This ensure the similar lengths of both blocks. Different EO-PO di-block copolymers have composition EO_{12}-PO_9, EO_{22}-PO_{17}, and EO_{44}-PO_{34}. The compositions of poloxamers used in this study were EO_{11}-PO_{16}-EO_{11}, PO_8-EO_{22}-PO_8, PO_{11}-EO_{27}-PO_{11} and PO_{14}-EO_{24}-PO_{14}, as provided by the producer.

The HPLC solvents (acetone, acetonitrile, methanol, and tetrahydrofuran) were purchased from (RCI Labscan, Thailand). Double deionized water was used for LC analysis by Milipore Progard 2 purification system. Mobile phases were mixed by volume and vacuum degassed.

2.2. Chromatography

Analysis of polymers at chromatographic critical conditions was performed at an Infinity 1200 series chromatograph by Agilent technologies, USA. A binary pump is connected with an auto-sampler followed by columns in a column oven (maintained at 30 °C unless otherwise stated) in a series. The eluent from the column is directed to an evaporative light scattering detector, ELSD (1200 series ELSD). Nitrogen was used as carrier gas at pressure of 50 psi while temperature of ELSD was maintained at 40 °C. For some experiments PL-ELS 2100 (Polymer Laboratories, Church Stretton, Shropshire, UK) or a SEDEX 45 ELSD (Sedere, France) was used with a modular system. Flow rate of mobile phase was kept 0.5 mL.min^{-1}. Sample solutions have concentration in range of 3.0 to 10 mg.mL^{-1}. The injection volume in all cases was 20 µL. The processing of the data was performed by Agilent Open LAB ChemStation for LC System.

Following columns are used in this study

- Prodigy ODS3: silica-based octadecyl phase; 250 × 4.6 mm; particle diameter = 5 µm; nominal pore size = 100 Å (Phenomenex, Torrance, CA, USA)
- Bakerbond C18: silica-based octadecyl phase; 250 × 4.6 mm; particle diameter = 5 µm; nominal pore size = 120 Å
- Jupiter RP-C18; 250 × 4.6 mm; pore size: 300 Å; particle diameter: 5 um (Phenomenex, USA).
- Nucleosil 500-7 C18:· silica-based octadecyl phase; 250 × 4 mm; particle diameter =7 µm; pore size = 500 Å (Macherey-Nagel, Germany)
- Nucleosil 4000-7 C18: silica-based octadecyl phase; 250 × 4 mm; particle diameter = 7 µm; pore size = 4000 Å. (Macherey-Nagel, Germany)

3. Results and discussion

As mentioned earlier, there are two options in liquid chromatography at critical conditions with regard to elution of non-critical block. The first option could be exclusion of non-critical block allowing elution of block copolymers earlier than the critical homopolymers, termed as LCCC-SEC. Secondly, the non-critical block may have strong interaction with the stationary phase allowing elution of block copolymers after the critical homopolymers, termed as LCCC-IC. Both situations can reveal important information within their limitations. Critical conditions of any polymer can vary among columns of different vendors or even the columns of different badges of the same vendor depending upon their carbon load. In this study, critical conditions are individually adjusted for each column that generally varied in a range of 5% of mobile phase composition. However, a single value is used for the discussed parameters in order to avoid complications in the mind of non-specialist readers.

3.1. LCCC-SEC

The elution of non-critical block in exclusion regime of LC of polymers in the most widely employed approach in analysis of block copolymers at critical conditions. LCCC-SEC approach is applicable to any size of polymer provided the excluded block is large enough to be separated well from the solvent peak. In this approach a calibration curve is constructed by analyzing homopolymers of non-critical block at critical conditions of the other block. This calibration curve is then used for estimation of individual block length of the non-critical block while separating critical homopolymers from the block copolymers. The approach allows fair estimation of individual block length and is literally the only method for estimation of individual block lengths of high polymers. In some cases, the length of critical block [25,26] and end-groups [51] may affect the elution of non-critical block, nonetheless, the method can be used for comparison of individual block lengths of non-critical block connected to critical block of similar length. Under these conditions, only practical variables that can be used for optimization of separation with respect to molar mass of excluded block are pore diameter and critical mobile phase selectivity.

3.1.1. Pore diameter

In size exclusion chromatography, separation is solely realized in the pore volume of the column. The elution of non-critical block in exclusion regime follows the same mechanism. The larger hydrodynamic volume of the excluded block results in earlier elution of the block copolymers compared to block copolymers with smaller hydrodynamic volume of excluded block. The elution volume of the block copolymer under current conditions is directly related to the molar mass of non-critical block. To demonstrate effect of pore diameter on the elution behavior, octadecyl columns with identical dimensions but different pore diameters are used. Chromatographic critical conditions of PMMA were realized on octadecyl columns in acetonitrile, tetrahydrofuran mobile phase system with 1% of additive trimethylamine to avoid unwanted polar interaction of free silanol groups with ether linkages of excluded PEO block of PEO-PMMA block copolymer. The polymers were synthesized by using PEO macro-initiators of varying molar masses while taking equal feed ratio of macro-initiator to monomer (MMA). In all cases, polymerization proceeded near to completion that is confirmed by NMR. Hence, the molar mass of individual blocks would be assumed similar. Nonetheless. In current case our major concern is molar mass of macro-initiator that is excluded from the stationary phase. Smaller pore diameter (120 **Å**) will allow better resolution of the block copolymers synthesized by low molar mass macro-initiator. A very nice separation of the block copolymers synthesized by macro-initiators having molar mass till 5000 g/mol was achieved. The resolution gets poor and actually no discrimination was possible for the products initiated with macro-initiators having molar mass above 5000 g/mol, Fig. 1A. The analysis of the same products at column of similar nature differing only in pore diameter (300 Å compared to 120 Å) under similar conditions otherwise is demonstrated in Fig. 1B. Not only the separation with respect to the short PEO-block (<5000 g/mol) is compromised, but in particular, the separation of the block copolymer with short PEO block from residual PMMA is barely achieved on 300 **Å** column. However, the separation of block copolymers initiated with macro-initiator of higher molar masses (between 5000 g/mol to 40,000 g/mol) is successfully achieved. The results clearly demonstrate the importance of selection of correct pore diameter for any LCCC-SEC analysis. Better resolution of low molar mass non-critical block can be achieved on smaller pore columns compared to larger pore columns that is more suitable for the polymers having larger non-critical block length. A multiple column approach using columns of different pore diameters in series similar to typical size exclusion chromatography is the option for enhancement of molar mass range of non-critical block in LCCC-SEC [52,53]. However, single column approach would be more practical in order to avoid requirement of extra solvent and time. In SEC, enhancement in the analyzable molar mass range using multiple column approach is practical since single calibrated system can be used for analysis of different kinds of polymers. This, however, is not the case in LCCC and critical conditions of every system has to be established separately. Generally, analyst know the molar mass range of the samples and it is more practical to use the single column that is capable of resolving required molar mass range of the excluded block.

3.1.2. Mobile phase composition

Preferential solvation and selectivity of critical mobile phase for hydrodynamic volume of non-critical block is another important factor in LCCC-SEC. In this context, chromatographic critical point of PMMA is established in acetone-water on the same 300 **Å** column. The behavior of two different critical mobile phases may be completely different with regard to solubility and hydrodynamic volume of the excluded block. Furthermore, the length of critical block may also influence the elution of block copolymer [25,26]. This effect was not very much pronounced in case of acetonitrile-THF mobile phase for PEO-PMMA block copolymers at CCP of PMMA and block copolymer eluted close to the elution volume of the non-critical macro-initiator. However, critical mobile phase composition of acetone-water resulted in significant effect of the attached critical block on the elution behavior of block copolymer. The deviation from the elution volume of macro-initiator increased with increase in the length of critical block, Fig. 2A. A dotted line in the plot shows the late elution of block copolymer compared to its parent macro-initiator (PEG_{20K}).

As a next step, the elution of block copolymer is evaluated while using same column but mobile phase composition that render exclusion of both PEO and PMMA. The block copolymer eluted earlier than both the macro-initiator and homopolymer of similar molar mass of PMMA as expected. On the same lines, elution of the same block copolymer at slight interaction conditions of PMMA while PEO is still in exclusion are shown Fig. 2B. In this case, both blocks pull the whole polymer in opposite direction that resulted in elution of block copolymer in between two precursors.

3.2. LCCC-IC

The applications of elution of non-critical block (and in turn the block copolymer) in interaction regime after critical point are generally limited to low molar mass block copolymers such as poloxamers (EO-PO block copolymers). This has to be a peculiar

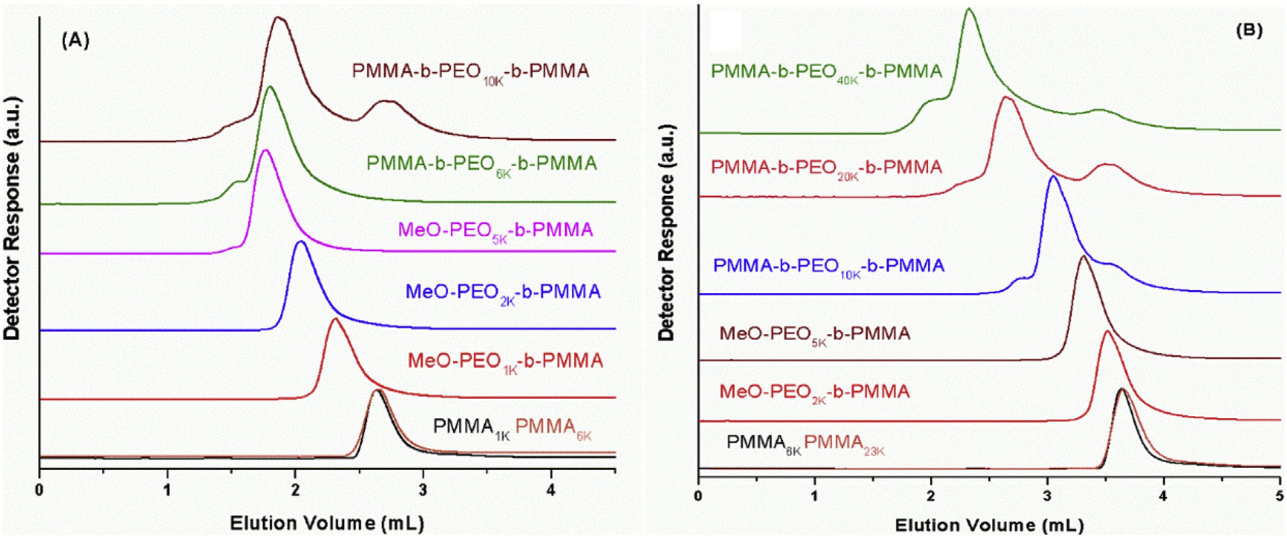

Fig. 1. LCCC-SEC of different di- and tri-block copolymers of PEO with PMMA having equal initial monomer to macro-initiator ratio (1:1) at CCP of PMMA [ACN/THF/TEA (v/v): 90/9/1 (v/v)]) on Ocatdecyl RP column, dimensions 250 x 4.6 mm, (A) pore size : 120 Å; (B) pore size 300 Å.

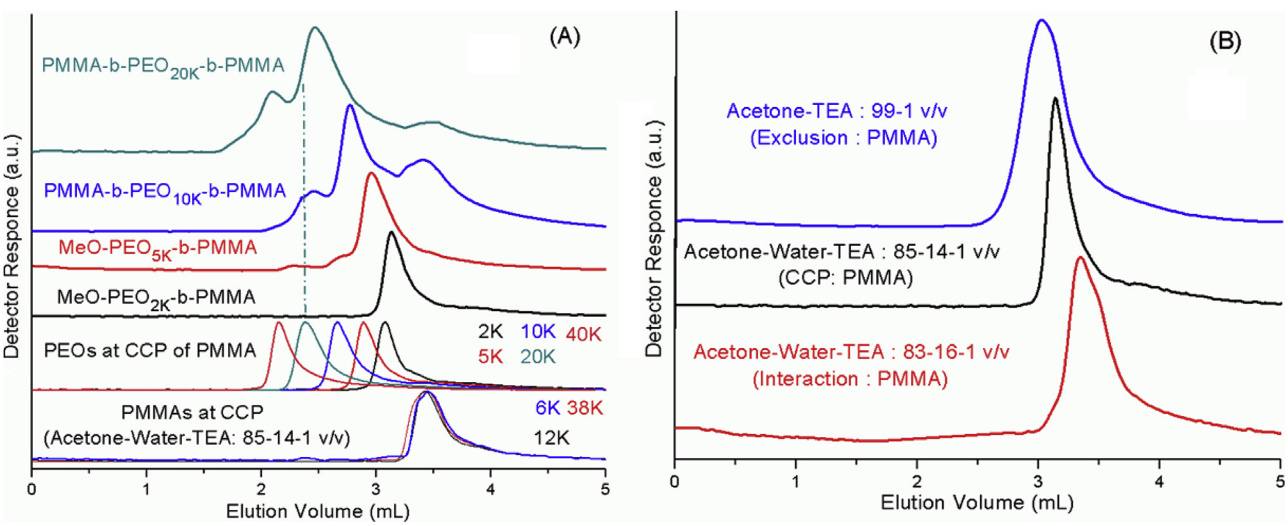

Fig. 2. (A)LCCC-SEC of different di- and tri-block copolymers of PEO with PMMA having equal initial monomer to macro-initiator ratio (1:1) at CCP of PMMA [Acetone/Water/TEA (v/v): 85/14/1 (v/v)]) on Ocatdecyl RP column, dimensions 250 x 4.6 mm, pore size 300 Å: (B) Elution behavior of MeO-PEO_{2K}-b-PMMA (1:1) as a function of mobile phase composition around CCP of PMMA (off-critical conditions).

situation in which the interaction of non-critical block with the stationary phase is fairly weak at critical point of the other block. LCCC-IC has been applied for oligomer separation of block copolymer with respect to non-critical block provided the interacting block is narrowly dispersed and have fairly weak interaction with the stationary phase [9,15,18,21,22,54–61]. The combination of critical point of one block while having reasonably low interaction of non-critical block is not easy to realize. A special case in this context is EO-PO di- and tri-block copolymers (poloxamers). There are several parameters that can be adjusted to improve the separation of the oligomers in LCCC-IC of poloxamers.

3.2.1. Pore diameter

Similar to LCCC-SEC, pore diameter is an important parameter that can influence the resolution of the oligomer peaks in LCCC-IC. The separation of block copolymers according to different block lengths of interacting block can be maneuvered by using columns of same nature but different porosity. Chromatographic critical point of any polymer on the columns of same nature and different porosity will be in the same range [18,20,62–65]. The individual block lengths of the block copolymers used in this study are similar since equal monomer to macro-initiator has been used for synthesis. Chromatographic critical point for PEO can be found on RP octadecyl column in an aqueous mobile phase containing 33% acetonitrile. A second chromatographic critical point of PEO on the same column is found in 59% acetonitrile in water. The change in mobile phase composition in between these two critical point does not induce any significant variation from critical behavior. Under these conditions, PPO interact with the stationary phase and elute later than the void volume of the column as separated oligomers. Therefore, EO-PO block copolymer should be separated with regard to PO repeat units under current conditions. A narrow pore column (**100 Å**) has shown an excellent separation of EO-PO block copolymers according to PO block having nine repeat units on average (EO_{12}-PO_9), Fig. 3A. The interaction of repeat units results in exponential increase in the elution volume hence the longer blocks would interact too strongly to elute on this column. The oligomer separation of EO_{12}-PO_9 with respect to PO block is compromised

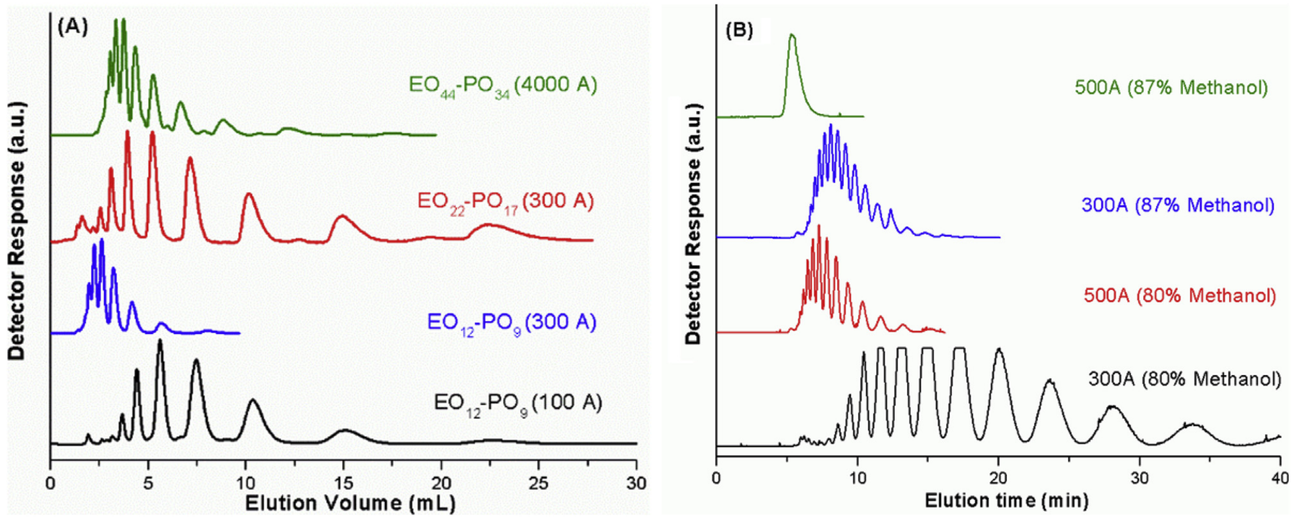

Fig. 3. (A) LCCC-IC of different EO-PO di-block copolymers having equal initial monomer to macro-initiator ratio (1:1) at CCP of PEO and interaction of PPO on Octadecyl columns having different pore diameter, mobile phase: acetonitrile-water (v/v): 38–62 (v/v); (B) LCCC-IC of commercial poloxamers (EO_{11}-PO_{16}-EO_{11}) at CCP of PEO and interaction of PPO on Octadecyl columns having different pore diameter in two different critical mobile phase compositions (1): methanol-water (v/v): 80-20 ; (2): methanol-water (v/v): 87–13.

while using a column with wide pore diameter (300 **Å**) under otherwise similar conditions. An approximate molar mass of 1000 g/mol for PO block of the product initiated by macro-initiator of same molar mass will have seventeen PO repeat units on average (EO_{22}-PO_{17}). This product was well separated with respect to oligomers of PO block by using octadecyl column having pore diameter of 300 **Å** under similar conditions otherwise. On the same lines the product initiated by MeO-PEO_{2K} would have thirty-four PO repeat units on average (EO_{44}-PO_{34}). A rational PO oligomer resolution of this product would only be possible on a column having even larger pore diameter under these conditions. The oligomer separation with respect to PO block of EO-PO block copolymer with a reasonable resolution was achieved by using an octadecyl column having pore diameter of 4000 Å, Fig. 3A.

Fig. 3B demonstrates separation of a commercial poloxamers (EO_{11}-PO_{16}-EO_{11}) in methanol-water mobile phase. Similar to acetonitrile-water, PEO have two close critical points on octadecyl column in aqueous methanol mobile phase (80% and 87% methanol). The elution behavior of PEO remain fairly critical in between these two critical points. Pore diameter selectivity of oligomer separation of PPO at two different critical mobile phase compositions for PEO is depicted. An excellent baseline separation with regard to number of PO units in the block copolymer is achieved in mobile phase containing 80% methanol on a column having pore diameter of 300 **Å**. The separation of the same product is compromised especially for lower oligomers using same mobile phase (80% methanol) on a 500 **Å** column. The interaction strength of PPO reduced considerably while using critical mobile phase composition with higher methanol content (87%). The separation of the same product (EO_{11}-PO_{16}-EO_{11}) according to PO block using 87% methanol is not as good as was achieved by 80% methanol on 300 **Å** column. Furthermore, the oligomer separation of block copolymer according to PO block is completely lost in 87% methanol on a 500 **Å** column. The above discussion and the chromatograms shown in Fig. 3 clearly demonstrate the selectivity of pore diameter for the molar mass range of interacting block. Hence, a well-pondered selection of pore diameter is required for achieving better oligomer resolution of block copolymers with respect to interacting block at CCP of the other block. Multiple column approach in case of LCCC-IC will not be suitable due to exponential difference in the elution volume for successive oligomers on the stationary phase having same pore size. Furthermore, higher oligomers that are analyzable on a column with wide pores may retain irreversibly on the stationary phase having narrow pores that render multiple column approach impractical.

3.2.2. *Mobile phase composition*

Generally chromatographic critical point is a sharp point in a special mobile phase composition and temperature for any polymer on any particular column. Any change in the mobile phase composition or temperature results in strong deviation from ideal behavior either in exclusion or interaction regime. On the other hand, in aqueous acetonitrile and aqueous methanol mobile phases on RP octadecyl columns, PEO have two close critical points [17,18]. In acetonitrile-water mobile phase, first critical point is achieved at 33% acetonitrile while second at 59% acetonitrile. On the same line, first critical point of PEO on RP octadecyl columns is realized in 80% methanol while second in 87% methanol in water. For both cases, elution behavior of PEO remain close to critical while using any mobile phase composition between two critical points. Moreover, the strength of interaction of interacting block may vary while mobile phase composition is changed. Fig. 4A demonstrates the oligomer separation of block copolymer according PO block of EO-PO di-block copolymers synthesized by using MeO-PEO of different molar masses while using equal monomer to macro-initiator feed ratio on a **100 Å** column. A baseline separation of block copolymer according to PO block is achieved in mobile phase containing 33% acetonitrile for EO_{12}-PO_9. Using slightly stronger mobile phase composition (40% and 45% acetonitrile) resulted in earlier elution of the oligomers of the same product with overlapped peaks. Further increase in acetonitrile content to 53% allowed for separation of EO_{22}-PO_{17} with respect to PO block. The oligomer separation EO_{22}-PO_{17} is further compromised while using mobile phase containing 59% acetonitrile. The effect of mobile phase composition on oligomer separation of block copolymer according to PO block while using mobile phase composition between two critical points in methanol-water for EO_{11}-PO_{16}-EO_{11} is demonstrated on a 500 **Å** column, Fig. 4B. An excellent oligomer separation achieved by 78% methanol get compromised slowly by increasing methanol content in the mobile phase leading to a single broad peak while using a mobile phase consisting of 87% methanol in water. The broadness of peaks

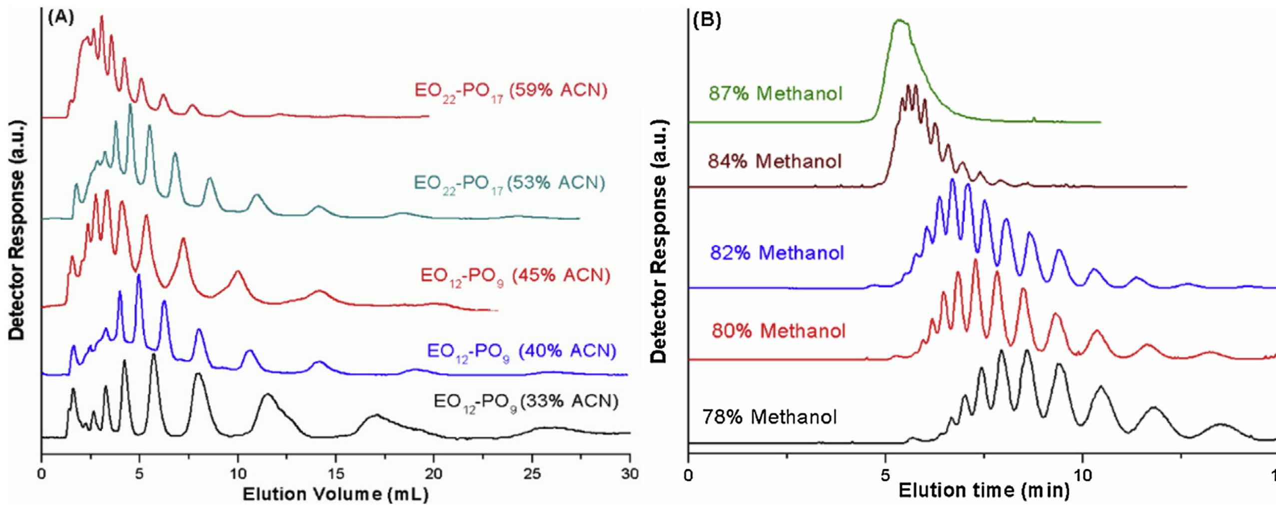

Fig. 4. (A) LCCC-IC of different EO-PO di-block copolymers having equal initial monomer to macro-initiator ratio (1:1) at CCP of PEO and interaction of PPO on Octadecyl 100 Å column using different mobile phase composition between two critical points at 33% and 59% (v/v) aqueous acetonitrile where PEOs exhibit near critical behavior while interaction strength of PPO is significantly affected; (B) LCCC-IC of EO_{11}-PO_{16}-EO_{11} at CCP of PEO on Octadecyl 500 Å column using different mobile phase composition between two critical points at 78% and 87% (v/v) aqueous methanol where PEOs exhibit near critical behavior while interaction strength of PPO is significantly affected.

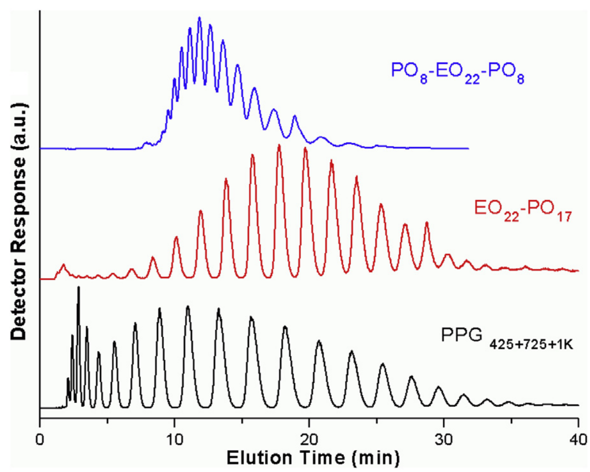

Fig. 5. Linear binary solvent gradient of methanol-water (80–90% methanol in 30 min) close to critical conditions of PEO on octadecyl RP 100 Å column.

in 78% methanol may be attributed to weak interaction of PEO with the stationary phase. Hence, the selection of suitable mobile phase composition according to the length of interacting PO block is imperative while keeping PEO fairly close to critical conditions.

3.2.3. Binary solvent gradient close to critical conditions

The existence of a fairly large mobile phase composition without much effect on the critical behavior allows running a gradient in mobile phase composition rendering near critical conditions. This type of typical behavior exists in aqueous mobile phases of methanol and acetonitrile on RP columns. Variations in the mobile phase composition in fairly near critical conditions of PEO render an influence on the retention of PPO that allows running a gradient for improvement of resolution of block copolymer according to interacting PPO oligomers. Fig. 5 demonstrates an excellent oligomer separation of a mixture of PPGs of fairly broad molar mass range that is not possible by using any isocratic system. Similar baseline oligomer separation of EO_{22}-PO_{17} is realized under same conditions. In principle, separation in LCCC-SEC should be very weakly influenced architecture or arrangement of the blocks [9,56]. A-B-A and B-A-B tri-block copolymers should show similar behavior at critical conditions of either block in LCCC-SEC. Conversely, separation in LCCC-IC is strongly influenced by the architecture and resolution with respect to interacting PO block depend upon the position of critical block in the block copolymer. The elution behavior of interacting homopolymers B, AB di-block copolymers and A-B-A tri-block copolymers should be comparable at chromatographic critical point of A [17,18,66]. This, however, is not the case for B-A-B tri-block copolymers analyzed under similar conditions [61,66,67]. EO-PO di-block and PO-EO-PO tri-block copolymers shown in Fig. 5 have similar number of average PO repeat units. However, the separation is drastically influenced by the presence of critical block in the center due to different topology [68]. The variable length of the critical block in center also contribute by hindering the interaction of the side blocks. The synergistic effect of above-mentioned factors lead to incomplete separation and early elution of block copolymer with respect to PO oligomers. Hence, the full separation of PO oligomers of PO-EO-PO tri-block copolymers requires weaker mobile phase for PPO in the mobile phase composition where critical behavior of PEO is not significant effected. This can be realized by using a gradient of two critical mobile phases for PEO having considerably different strength for interacting PPO block.

3.2.4. Gradient between two critical mobile phase compositions

Chromatographic critical conditions of PEO in two different mobile phase compositions should have different strength for the PO block. CCP of PEO can be realized in acetone-water having 37% acetone. Under these conditions PPO strongly interacts with the stationary phase allowing elution of only very low oligomers [17,18]. On the same column, CCP of PEO can be realized in 87% methanol that renders an excellent separation of PPG_{1K} within fifteen minutes. However, an insufficient oligomer separation of a commercial poloxamer having formula EO_{11}–PO_{16}-EO_{11} could be achieved, Fig. 6. The separation of EO_{11}-PO_{16}-EO_{11} with regard to PO oligomers can be improved by using a gradient of above-mentioned critical mobile phases [17,69,70]. It is pertinent to mention here that gradient of two critical condition does not strictly ensure the critical behavior throughout, nonetheless, it remain fairly close to critical conditions [69–71]. The critical mobile phase for PEO containing 37% acetone is designated a “solvent A” while 87% methanol is designated as “solvent B” in the proceeding text. The gradient can be maneuvered as per the length of the inter-

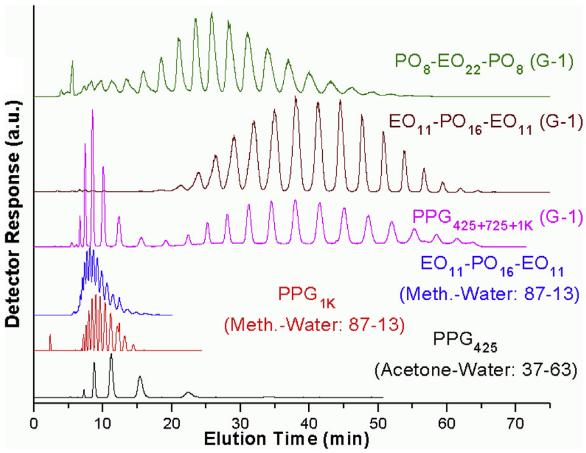

Fig. 6. Separation of PPGs of different molar masses in critical mobile phase compositions for PEO, acetone-water (v/v): 37–63 (solvent A) and methanol-water (v/v): 87–13 (solvent-B), Separation of EO_{11}-PO_{16}-EO_{11} in solvent B, Separation of mixture of PPGs of different molar masses along with commercial poloxamers having similar average number of PO repeat units but position of the critical block EO_{11}-PO_{16}-EO_{11}, and PO_8-EO_{22}-PO_8 using gradient of two critical mobile phase compositions. Gradient-1 starts at 50% B, in 15 min reaches to 75% B, to 100% B in 60 min, and maintained at 100% B till 70 min.

acting PPO block. First gradient (G-1) of solvent A and solvent B is designed to improve the separation of tri-block copolymers of both A-B-A and B-A-B type containing sixteen PO units on average. The initial composition of G-1 was 50% of solvent A and 50% solvent B. The content of solvent B is increased to 75% in fifteen minutes and further to 100% till sixty minutes of the runtime. The composition of mobile phase was maintained at 100% B till 70 min before switching to initial mobile phase composition (50% mixture A and B) for the next run. A mixture of PPG 425, 725 and 1 K is excellently separated using G-1. Furthermore, G-1 allows an excellent baseline separation of tri-block copolymers having approximately sixteen PO repeat units. It is pertinent to mention here that although baseline separation of critical EO block in the center has been achieved in the gradient, the elution volumes are considerably less than the EO-PO tri-block copolymer with interacting PO block in the center having otherwise similar composition. The earlier elution of PO_8-EO_{22}-PO_8 compared to EO_{11}-PO_{16}-EO_{11} may be attributed to presence of the critical block in the center [61,66,67]. The described gradient (G-1) allows for only incomplete oligomer separation of the products containing more than twenty PO repeat units on average. These products would require adjusting the gradient steepness and strength at different points of time. In this context, gradient-2 (G-2) of two critical solvents is formulated in a such a way that it starts with higher initial solvent strength for PO and less steepness in the later half to have better resolution of higher oligomers. G-2 starts with 70% B, the composition reaches to 85% B in 30 min, followed by 100% B in 120 min. The composition was maintained at 100% B till 150 min. Fig. 7 demonstrates an excellent separation of a mixture of PPGs additionally containing PPG_{2K}. Excellent baseline separation was achieved for PPGs having repeat units below thirty. G-2 considerably improved separation of PO-EO-PO tri-block copolymers having 22 and 28 PO repeat units on average. Hence, gradient of the two critical solvents should be designed as per length of the interacting PO block. Lower oligomers require higher acetone-water (solvent A) for better resolution of block copolymer with regard to PO repeat units whereas higher initial content of methanol-water (solvent B) is required for better oligomer resolution of block copolymer containing comparatively longer PO block. On the same lines, consideration of topology of block copolymer is also important for selection of suitable gradient. Above demonstration of designing the gradient of two critical mobile phases

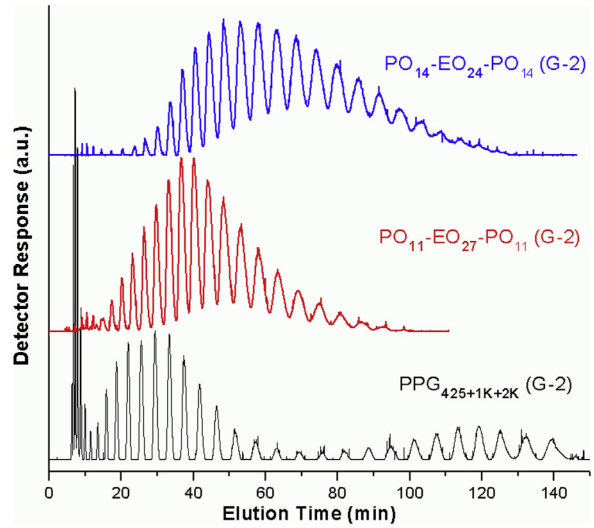

Fig. 7. Separation of mixture of PPGs, and poloxamers containing more than 20 PO repeat units (PO_{11}-EO_{27}-PO_{11} and PO_{14}-EO_{24}-PO_{14}) using gradient of two critical mobile phase compositions, acetone-water (v/v): 37–63 (solvent A) and methanol-water (v/v): 87–13. Gradient-2 (G-2) starts 70% B, in 30 min reaches to 85% B, to 100% B in 120 min, and maintained at 100% B till 150 min.

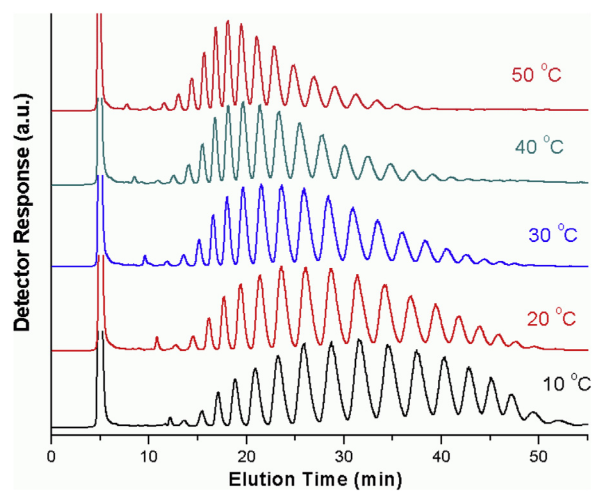

Fig. 8. Effect of temperature on elution behavior EO_{22}-PO_{17} by using a linear binary solvent gradient of methanol-water (80–90% methanol in 40 min) close to critical conditions of PEO on octadecyl RP 100 Å column.

clearly show pragmatic applicability of this approach for oligomer separation of commercial poloxamers and other related products.

3.2.5. *Temperature*

Temperature is often used for fine-tuning of critical point and for contriving the separation of interacting block of the block copolymers [37–42]. In most of the cases, chromatographic critical point is a sharp point with respect to mobile phase composition and temperature. However, existence of two close chromatographic critical points of PEO ensures their near critical behavior for the practical temperature range of LC separation. However, the interaction strength of the interacting block can be strongly influenced by changes in temperature. Fig. 8 demonstrates the effect of column oven temperature on oligomer separation of di-block EO-PO copolymer with respect to PO repeat units. The sharpness of peaks of block copolymer with respect to individual oligomers of PPO indicate a sufficient critical behavior of PEOs in the whole range of shown temperature. The analysis time has decreased from more than 50 min to around 35 min while doing the same chromatographic run at increased temperature from 10 °C to 50 °C. Furthermore, the peaks seem to be sharper and well-resolved at higher temperature compared to lower temperature.

4. Conclusion

In this study, we systematically demonstrate various parameters that can be adeptly selected to improve the separation while working at chromatographic critical conditions of polymers. Pore diameter of the column and critical mobile phase composition are the only important parameter that can be used to enhance separation efficiency in LCCC-SEC. Narrow pore columns are more suitable for low molar mass range of non-critical block whereas wide pore columns enhance the applicable individual block length of non-critical block. LCCC-SEC is applicable to a wide molar mass range of non-critical block. On the other hand, there are several experimental parameters that can be adeptly contrived to improve resolution in LCCC-IC. These important experimental variable include pore diameter of the column, mobile phase composition, binary solvent gradients close to critical conditions, gradient between two critical solvents that have different strength for interacting block and temperature. LCCC-IC is a very specific situation and can be applied to limited number of combinations of blocks having block lengths in oligomer range. Commercially most important polymers that can be analyzed by LCCC-IC are EO-PO di- and tri-block copolymers (called poloxamers). LCCC-SEC is generally not effected to large extent by the position of the critical block. However, the elution behavior is strongly influenced by the position of the critical block in case of LCCC-IC. Hence, the strength of this outstanding method can be immensely enhanced by careful selection of pore diameter of column, mobile phase composition, gradients, and temperature considering the length and position of the non-critical block.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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